

Christopher De Vault

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Recent B.S. Computer Science graduate with experience in machine learning, NLP, database systems, systems programming, and data analysis. Skilled in Python, SQL, pandas, scikit-learn, and relational database design with interests in analytics engineering and data-driven software systems.

Education

University of Arizona – Tucson, AZ

Bachelor of Science in Computer Science, Minor in Philosophy – May 2026

Relevant Coursework: Data Science, Database Design, Data Structures & Algorithms, OS, Systems Programming in C

Technical Skills

Languages: Python, SQL, Java, C, Unix

Data & Machine Learning: pandas, NumPy, scikit-learn, Matplotlib, NLP, BERT, PCA, TF-IDF

Databases & Analytics: Oracle SQL, JDBC, Relational Databases, Data Modeling

Systems & Developer Tools: Git, Github, Linux, Docker, Bash, UTM,

Concepts: Machine Learning, Data Visualization, Operating Systems, Systems Programming, Database Design, Testing

Projects

Large-Scale NLP Classification & Feature Engineering Pipeline – Python, pandas, scikit-learn, BERT, NLP

- Built an end-to-end NLP classification pipeline to analyze ~20,000 labeled social media posts using TF-IDF vectorization and transformer embeddings.
- Engineered high-dimensional feature representations using 768-dimensional BERT embeddings and PCA-based dimensionality reduction.
- Trained and evaluated Logistic Regression and SVM classifiers, achieving approximately 0.80 weighted F1-score on unseen test data.

Data Warehouse & Relational Database Engineering Project – Java, SQL, Oracle, JDBC, Data Modeling, Git

- Designed and implemented a normalized Oracle relational database containing 15+ interconnected entities and complex business relationships.
- Developed advanced analytical SQL queries using multi-table joins, aggregation, grouping, and filtering to generate operational insights.
- Performed formal functional dependency analysis and normalization through BCNF to improve consistency and reduce redundancy.

Virtual Memory Simulator & Page Replacement Analysis – Operating Systems, C, Linux, Performance Analysis

- Developed a virtual memory simulator implementing OPT, CLOCK, Rand, and NRU page replacement algorithms using configurable frame allocation and single-level page tables.
- Analyzed over 2.4 million Valgrind memory trace accesses to benchmark page fault behavior, dirty page writebacks, and workload locality characteristics.
- Conducted comparative performance analysis of replacement policies using quantitative metrics including fault ratios, disk-write frequency, and memory efficiency tradeoffs.

Expression Parser & Interpreter in Go – Go, Parsing, ASTs, Compiler Concepts

- Built a custom arithmetic expression interpreter in Go featuring lexical analysis, parsing, AST generation, and expression evaluation.
- Designed modular compiler-style components including tokenization, recursive descent parsing, and runtime execution logic.
- Implemented REPL functionality and structured error handling for interactive expression evaluation.

Experience

Western Regional Collegiate Cyber Defense Competition (WRCCDC) University of Arizona Cyber Saguaro

- Represented the University of Arizona in a regional cybersecurity competition involving 27 universities
- Led inject response operations, producing technical reports and remediation plans accounting for approximately 50% of total team scoring.
- Prioritized operational tasks under time pressure while maintaining system availability during active attacks